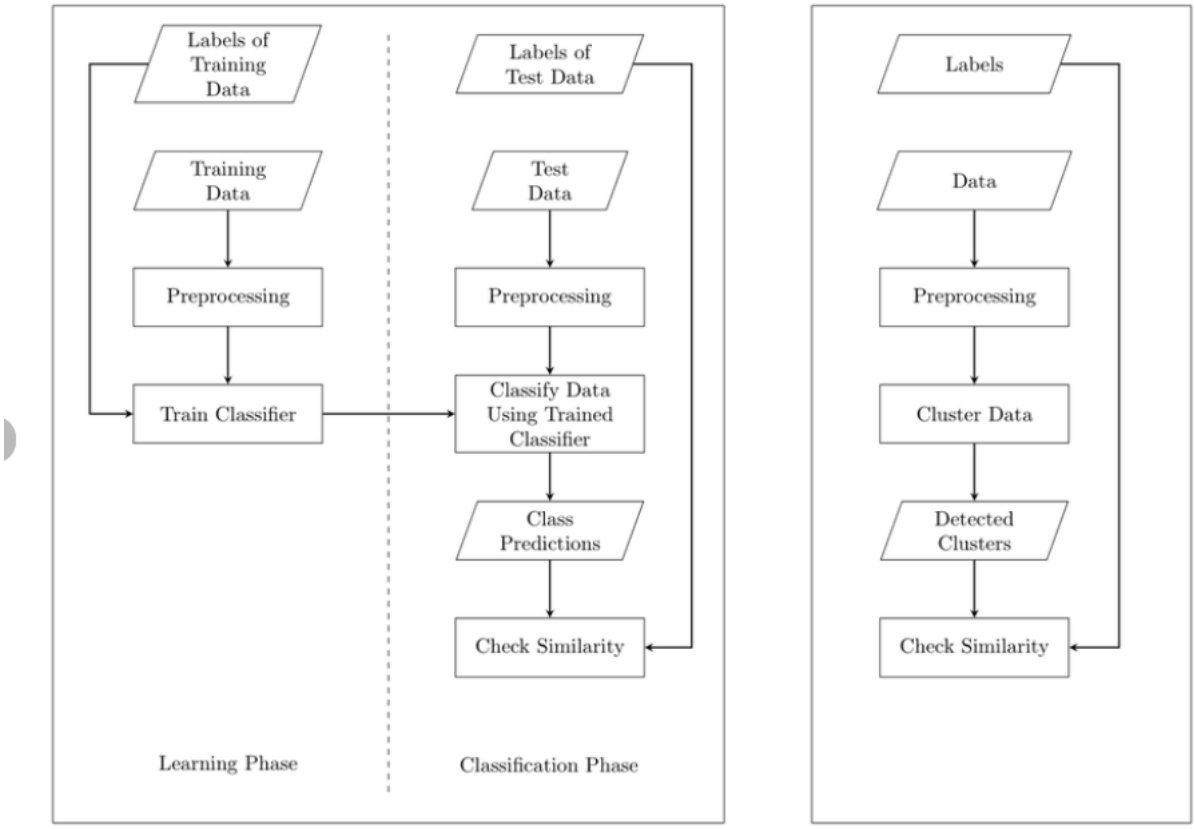
Topic: Unsupervised Learning & K-Means Clustering

Date: March 18, 2026

1. What is Unsupervised Learning?

Unsupervised Learning is a branch of machine learning where the model works with **unlabeled data**. Unlike supervised learning, there is no supervisor or "correct answer" provided. The model must independently discover the inherent structure, patterns, or groupings within the input data.



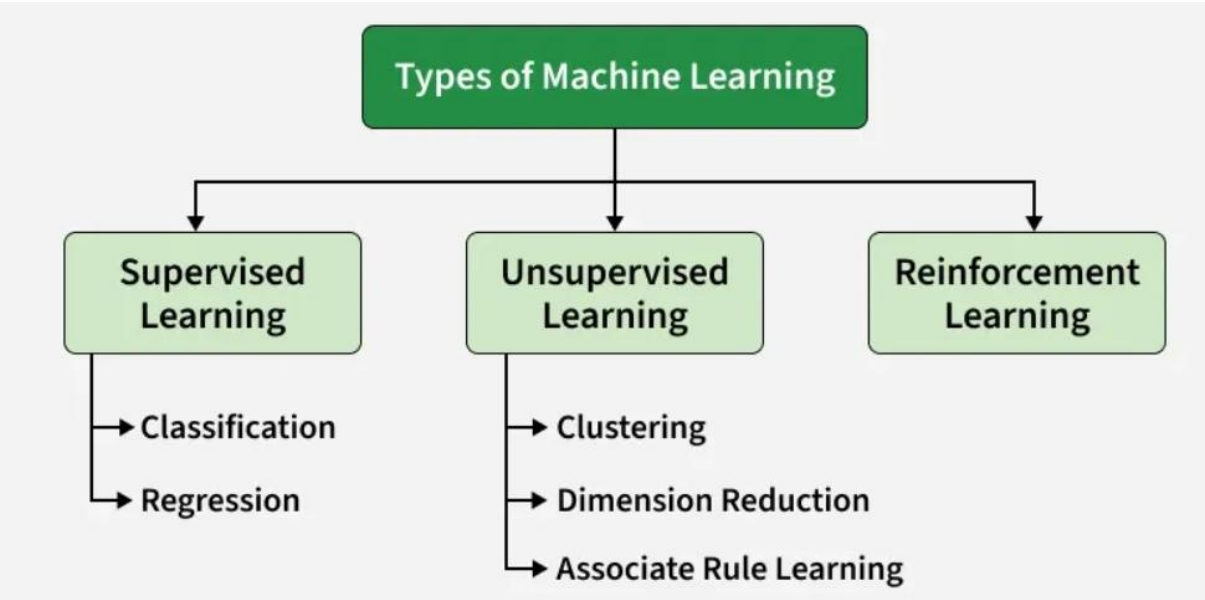
(a) Supervised Techniques (b) Unsupervised Techniques

2. Types of Unsupervised Learning

Unsupervised learning is typically divided into three main categories:

- **Clustering:** Grouping data points so that objects in the same group are more similar to each other than to those in other groups.
- **Association:** Discovering rules that describe your data, such as "customers who buy X also tend to buy Y."

- **Dimensionality Reduction:** Simplifying data by reducing the number of input variables while keeping the most important information.

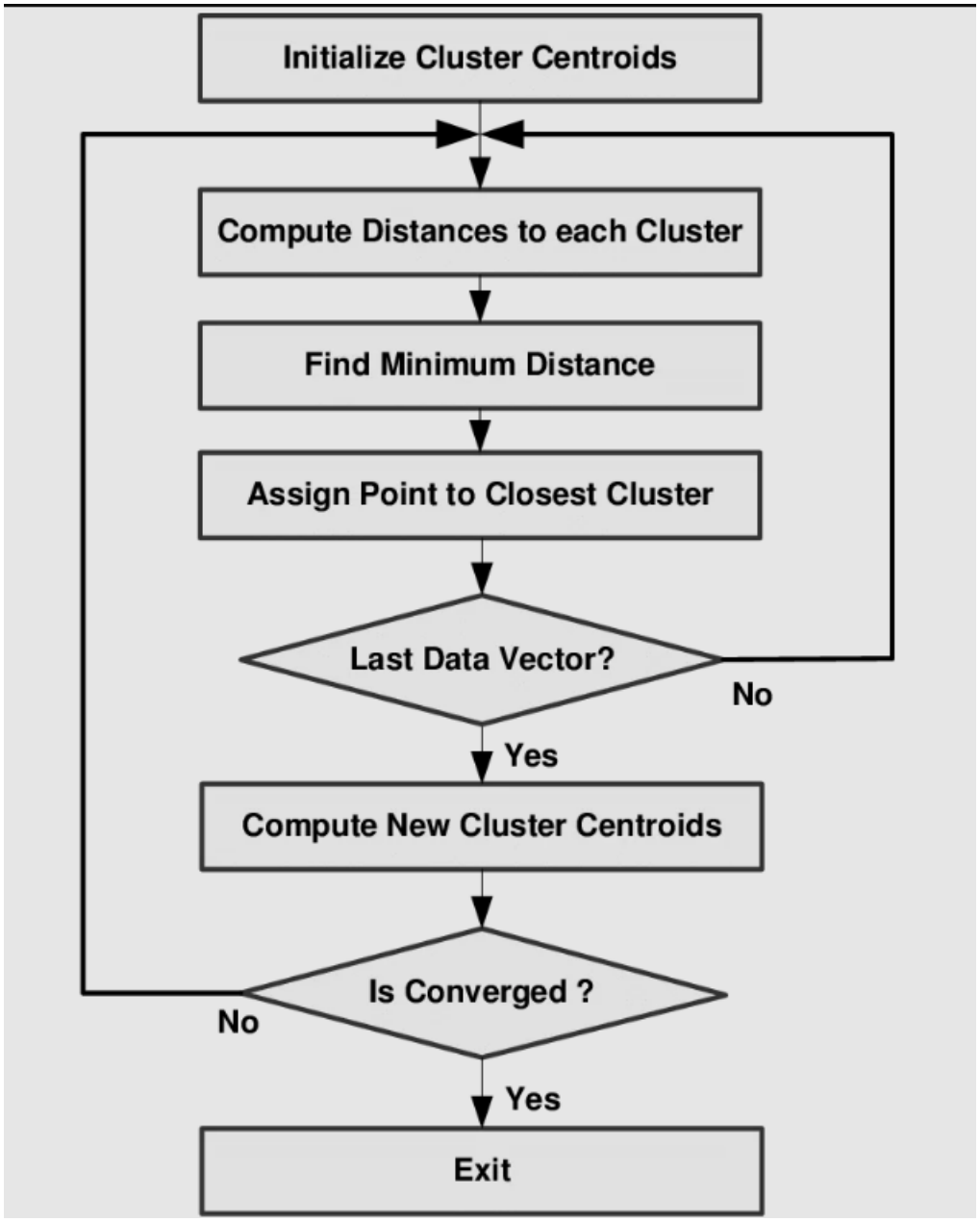


3. K-Means Clustering

K-Means is an iterative algorithm that partitions a dataset into **K** pre-defined, non-overlapping subgroups (clusters). It works by minimizing the distance between data points and the center of their assigned cluster (the centroid).

The Process:

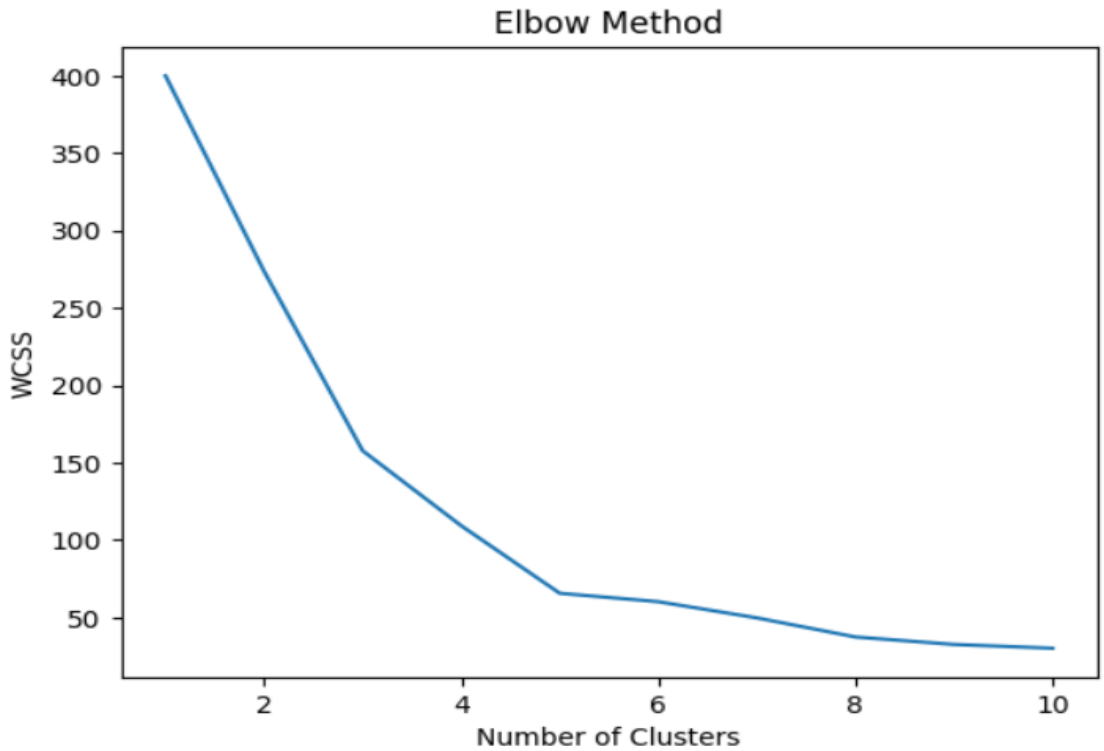
1. Select the number of clusters (K).
2. Randomly initialize K centroids.
3. Assign each data point to the nearest centroid.
4. Calculate the new mean (center) of each cluster and move the centroid.
5. Repeat until the centroids no longer change.

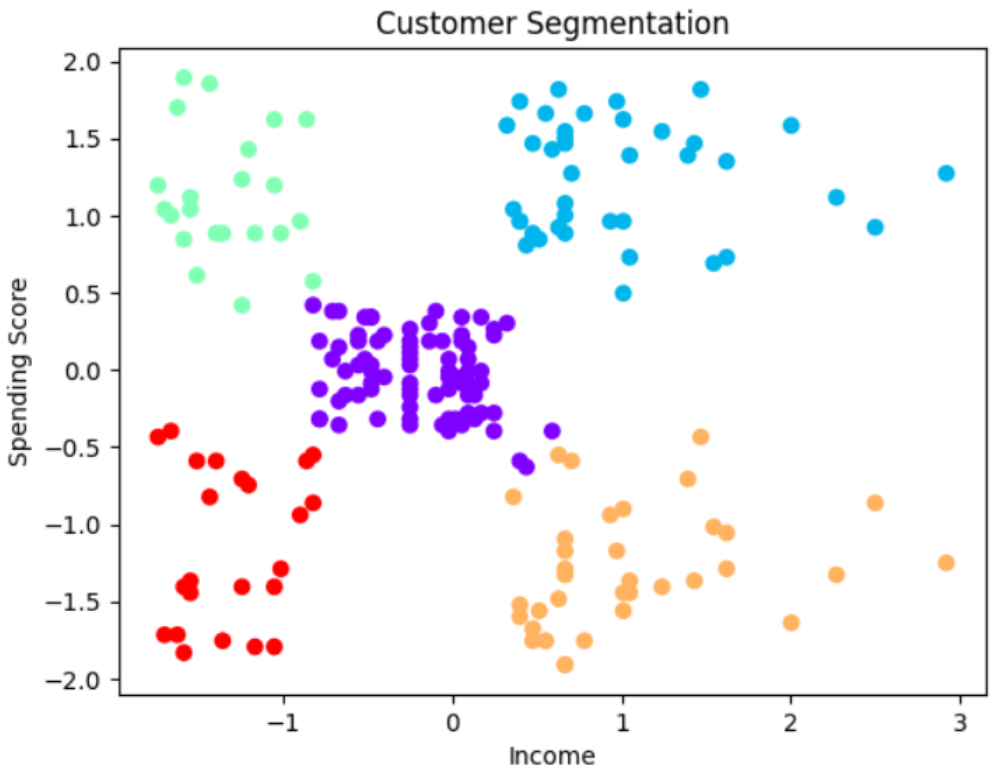


4. Graphs Used in Analysis

The Elbow Method

This line graph is used to determine the optimal number of clusters. It plots the **Within-Cluster Sum of Squares (WCSS)** against the number of clusters. The "elbow" of the curve indicates the point where adding more clusters does not provide significantly better modeling of the data.



Cluster Scatter Plot

Once the model is trained, a scatter plot visualizes the final groups. Each color represents a different cluster, and the **Centroids** (the mathematical centers) are often marked with a distinct symbol to show how the algorithm has partitioned the space.